

B K Birla Institute of Engineering & Technology, Pilani

45 Days Summer Training Program

Subject: Data Science with AIML

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Assignment 5

Section A – NumPy

Question 1:

Program:

```
import numpy as np
```

```
arr = np.random.randint(1, 101, 20)
```

```
print("Array:")
```

```
print(arr)
```

```
print("Minimum =", np.min(arr))
```

```
print("Maximum =", np.max(arr))
```

```
print("Mean =", np.mean(arr))
```

```
print("Standard Deviation =", np.std(arr))
```

Output:

Array:

```
[45 72 19 81 36 54 67 29 91 10 58 76 88 23 64 31 95 41 52 79]
```

Minimum = 10

Maximum = 95

Mean = 55.55

Standard Deviation = 24.42

Question 2:

Program:

```
import numpy as np

matrix = np.array([
    [5, 8, 9, 2],
    [6, 7, 1, 4],
    [3, 12, 15, 10],
    [11, 14, 13, 16]
])

print("Original Matrix:")
print(matrix)

print("\nDiagonal Elements:")
print(np.diag(matrix))

new_matrix = matrix.copy()
new_matrix[new_matrix % 2 == 0] = 0

print("\nMatrix after replacing even numbers with 0:")
print(new_matrix)

print("\nIndex of Maximum Value:")
print(np.argmax(matrix))
```

Output:

Original Matrix:

```
[[ 5  8  9  2]
 [ 6  7  1  4]
 [ 3 12 15 10]
 [11 14 13 16]]
```

Diagonal Elements:

```
[ 5  7 15 16]
```

Matrix after replacing even numbers with 0:

```
[[ 5  0  9  0]
 [ 0  7  1  0]
 [ 3  0 15  0]
 [11  0 13  0]]
```

Index of Maximum Value:

15

Question 3:

Program:

```
import numpy as np
```

```
A = np.array([
    [1,2,3],
    [4,5,6],
    [7,8,9]
])
```

```
B = np.array([
    [9,8,7],
    [6,5,4],
    [3,2,1]
])
```

```
print("First Matrix:")
print(A)
```

```
print("\nSecond Matrix:")
print(B)
```

```
print("\nAddition:")
print(A + B)
```

```
print("\nSubtraction:")
print(A - B)
```

```
print("\nMatrix Multiplication:")
print(np.matmul(A, B))
```

Output:

First Matrix:

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Second Matrix:

```
[[9 8 7]
 [6 5 4]
 [3 2 1]]
```

Addition:

```
[[10 10 10]
 [10 10 10]
 [10 10 10]]
```

Subtraction:

```
[[ -8 -6 -4]
 [ -2  0  2]
 [  4  6  8]]
```

Matrix Multiplication:

```
[[ 30 24 18]
 [ 84 69 54]
 [138 114 90]]
```

Question 4:

Program:

```
import numpy as np

a = np.linspace(1, 5, 5)
b = np.zeros(5)
c = np.ones(5)

print("Array using linspace:")
print(a)

print("\nArray using zeros:")
print(b)

print("\nArray using ones:")
print(c)

result = np.vstack((a, b, c))

print("\nFinal Stacked Array:")
```

```
print(result)
```

Output:

Array using linspace:

```
[1. 2. 3. 4. 5.]
```

Array using zeros:

```
[0. 0. 0. 0. 0.]
```

Array using ones:

```
[1. 1. 1. 1. 1.]
```

Final Stacked Array:

```
[[1. 2. 3. 4. 5.]
```

```
 [0. 0. 0. 0. 0.]
```

```
 [1. 1. 1. 1. 1.]]
```

Section B – Pandas

Question 5:

Program:

```
import pandas as pd
```

```
data = {  
    "Name": ["Aman", "Riya", "Karan", "Priya", "Rahul", "Neha", "Arjun", "Sneha", "Rohit", "Pooja"],  
    "Age": [20, 21, 19, 22, 20, 21, 23, 20, 22, 19],  
    "Marks": [82, 75, 91, 68, 88, 79, 95, 73, 84, 77],  
    "City":  
    ["Delhi", "Jaipur", "Mumbai", "Pune", "Delhi", "Chennai", "Kolkata", "Lucknow", "Bhopal", "Indore"]  
}
```

```
df = pd.DataFrame(data)
```

```
print("First 5 Rows:")
```

```
print(df.head())
```

```
print("\nBasic Statistics:")
```

```
print(df.describe())
```

```
print("\nStudents Scoring More Than 75 Marks:")  
print(df[df["Marks"] > 75])
```

```
print("\nSorted by Marks (Descending):")  
print(df.sort_values(by="Marks", ascending=False))
```

Output:

First 5 Rows:

	Name	Age	Marks	City
0	Aman	20	82	Delhi
1	Riya	21	75	Jaipur
2	Karan	19	91	Mumbai
3	Priya	22	68	Pune
4	Rahul	20	88	Delhi

Basic Statistics:

	Age	Marks
count	10.000000	10.000000
mean	20.700000	81.200000
std	1.251666	8.280499
min	19.000000	68.000000
25%	20.000000	75.500000
50%	20.500000	80.500000
75%	21.750000	86.500000
max	23.000000	95.000000

Students Scoring More Than 75 Marks:

Aman, Karan, Rahul, Neha, Arjun, Rohit, Pooja

Sorted by Marks (Descending):

Arjun
Karan
Rahul
Rohit
Aman
Neha
Pooja
Riya
Sneha
Priya

Question 6:

Program:

```
import pandas as pd
import numpy as np

data = {
    "Name":["Aman","Riya","Karan","Priya","Rahul"],
    "Marks":[85,np.nan,90,np.nan,78],
    "City":["Delhi","Mumbai",np.nan,"Jaipur","Pune"]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

print("\nMissing Values:")
print(df.isnull())

mean_marks = df["Marks"].mean()
df["Marks"] = df["Marks"].fillna(mean_marks)

print("\nAfter Filling Missing Marks:")
print(df)

df = df.dropna()

print("\nAfter Dropping Missing City:")
print(df)
```

Output:

Original DataFrame:

	Name	Marks	City
0	Aman	85.0	Delhi
1	Riya	NaN	Mumbai
2	Karan	90.0	NaN
3	Priya	NaN	Jaipur
4	Rahul	78.0	Pune

Missing Values:

(True/False table displayed)

After Filling Missing Marks:
Missing marks replaced with mean value.

After Dropping Missing City:
Rows having missing City are removed.

Question 7:

Program:

```
import pandas as pd
```

```
data = {  
    "Product":["Laptop","Phone","Tablet","Mouse","Keyboard"],  
    "Category":["Electronics","Electronics","Electronics","Accessories","Accessories"],  
    "Sales":[50000,35000,20000,5000,7000],  
    "Region":["North","South","East","West","North"]  
}
```

```
df = pd.DataFrame(data)
```

```
df.to_csv("sales.csv", index=False)
```

```
sales = pd.read_csv("sales.csv")
```

```
print("Sales Data:")
```

```
print(sales)
```

```
print("\nCategory Wise Total Sales:")
```

```
print(sales.groupby("Category").agg({"Sales":"sum"}))
```

```
print("\nTop 3 Products:")
```

```
print(sales.sort_values(by="Sales", ascending=False).head(3))
```

Output:

Category Wise Total Sales:

Accessories 12000

Electronics 105000

Top 3 Products:

Laptop
Phone
Tablet

Question 8:

Program:

```
import pandas as pd

df1 = pd.DataFrame({
    "StudentID":[101,102,103,104],
    "Name":["Aman","Riya","Karan","Priya"],
    "Department":["CSE","ECE","ME","CSE"]
})

df2 = pd.DataFrame({
    "StudentID":[101,102,103,104],
    "Marks":[91,74,87,95],
    "Grade":["A","B","A","A"]
})

merged = pd.merge(df1, df2, on="StudentID")

print("Merged DataFrame:")
print(merged)

print("\nStudents with Grade A:")
print(merged[merged["Grade"]=="A"])
```

Output:

Merged DataFrame:

	StudentID	Name	Department	Marks	Grade
0	101	Aman	CSE	91	A
1	102	Riya	ECE	74	B
2	103	Karan	ME	87	A
3	104	Priya	CSE	95	A

Students with Grade A:

Aman

Karan
Priya

Section C – Data Visualization

Question 9:

Program:

```
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English", "Computer", "Physics"]
marks = [85, 78, 90, 88, 92]

students = ["Aman", "Riya", "Karan", "Priya", "Rahul"]
student_marks = [82, 76, 91, 88, 95]

# Line Plot
plt.figure(figsize=(7, 5))
plt.plot(subjects, marks, marker="o", linewidth=2, label="Marks")
plt.title("Marks Trend Across Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.legend()
plt.grid(True)
plt.show()

# Bar Chart
plt.figure(figsize=(7, 5))
plt.bar(students, student_marks, label="Marks")
plt.title("Marks of Five Students")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.legend()
plt.grid(axis="y")
plt.show()
```

Question 10:

Program:

```

import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Math": [78, 85, 91, 88, 95],
    "Science": [82, 89, 86, 90, 94],
    "English": [75, 80, 78, 84, 87],
    "Computer": [90, 93, 88, 95, 97]
}

df = pd.DataFrame(data)

# Histogram
df.hist(figsize=(8, 6))
plt.suptitle("Histogram of Subject Marks")
plt.show()

# Box Plot
df.plot(kind="box", figsize=(6, 5))
plt.title("Box Plot of Subject Marks")
plt.grid(True)
plt.show()

# Pie Chart
grades = pd.Series(
    [12, 10, 8, 5],
    index=["A", "B", "C", "D"]
)

grades.plot(kind="pie", autopct="%1.1f%%", startangle=90)
plt.title("Grade Distribution")
plt.ylabel("")
plt.show()

```

Question 11:

Program:

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

```

```

np.random.seed(10)

data = {
    "Age": np.random.randint(22, 45, 50),
    "Salary": np.random.randint(30000, 90000, 50),
    "Department": np.random.choice(["HR", "IT", "Sales"], 50),
    "Experience": np.random.randint(1, 15, 50)
}

df = pd.DataFrame(data)

# Scatter Plot
plt.figure(figsize=(7, 5))
sns.scatterplot(data=df, x="Age", y="Salary", hue="Department")
plt.title("Age vs Salary")
plt.show()

# Heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(df[["Age", "Salary", "Experience"]].corr(),
            annot=True,
            cmap="viridis")
plt.title("Correlation Heatmap")
plt.show()

# Bar Plot
plt.figure(figsize=(6, 4))
sns.barplot(data=df, x="Department", y="Salary")
plt.title("Average Salary by Department")
plt.show()

```

Question 12:

Program:

```

import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [15, 25, 20, 30, 35]

plt.figure(figsize=(10, 8))

# Line Plot

```

```
plt.subplot(2, 2, 1)
plt.plot(x, y, marker="o")
plt.title("Line Plot")
```

```
# Scatter Plot
plt.subplot(2, 2, 2)
plt.scatter(x, y)
plt.title("Scatter Plot")
```

```
# Bar Chart
plt.subplot(2, 2, 3)
plt.bar(x, y)
plt.title("Bar Chart")
```

```
# Histogram
plt.subplot(2, 2, 4)
plt.hist(y, bins=5)
plt.title("Histogram")
```

```
plt.suptitle("Matplotlib Subplots")
plt.tight_layout()
```

```
plt.show()
```